CHAPTER 1

Degree QUDs and focus-sensitive scalar particles

Chinese dōu, cái, and jiù

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THIS chapter develops a degree-QUD-based analysis of Chinese focus-sensitive scalar particles (e.g., *dōu*, *cái*, *jiù*), using **norm sensitivity** (**mirativity**) as a lens. At first glance, these particles fall into two classes: (i) those akin to English *even* or *already*, giving rise to an **above-norm** inference (see (1)), and (ii) those akin to English *only* or German *erst*, giving rise to a **below-norm** inference (see (2)).

(1) **Particles akin to English *even*, *already*: e.g., *dōu*, *hái*, *lián...yě***

a. tā **dōu** [sān]$_F$ suì le
3SG EVEN three year SFP
'She is already 3 years old.' ⇝ She is older than expected

b. (**lián**) [Zhāng-Sān]$_F$ **dōu** lái-le
(EVEN) Zhāng-Sān EVEN come-PRF/SFP
'Even Zhāng Sān came.' ⇝ ZS's coming is beyond expectations

c. **lián** [Zhāng-Sān]$_F$ **yě** lái-le
(EVEN) Zhāng-Sān ALSO come-PRF/SFP
'Even Zhāng Sān came too.' ⇝ ZS's coming is beyond expectations

d. Zhāng-Sān bǐ [hóu-zi]$_F$ **hái** mǐnjié
Zhāng-Sān COMPARE monkey EVEN agile
'Zhāng Sān is even more agile than a monkey.'
⇝ Both ZS and monkeys are above the threshold of being agile

(2) **Particles akin to English *only* or German *erst*: *jiù*, *cái*, *zhǐ***

a. tā { **jiù** / **cái** / **zhǐ** yǒu } [sān]$_F$ suì
3SG JUST JUST ONLY have/EXIST three year
'She is only/just 3 years old.' ⇝ She is not as old as expected

b. Zhāng-Sān { **jiù** / **zhǐ** } chī-le [píng-guǒ]$_F$
Zhāng-Sān JUST ONLY eat-PRF apple(s)
'Zhāng Sān only ate apples.' ⇝ What ZS ate is below expectations

c. { **<u>jiù</u>** / **<u>zhǐ</u>** yǒu } [Zhāng-Sān]$_F$ lái-le
JUST ONLY have/EXIST Zhāng-Sān come-PRF/SFP
'<u>Only</u> Zhāng Sān came.' ⇝ Attendance was <u>below expectations</u>

d. tā (**zhì-shǎo**) huā-le [sān]$_F$ tiān **<u>cái</u>** kàn-wán [yī]$_F$ běn shū
3SG (at-least) take-PRF three day read-finish one CL book
'It took her <u>no less than</u> three days to finish one book.' (cf. (2a), (2e))
<u>More</u> time was spent <u>than expected</u>; <u>less</u> was accomplished <u>than expected</u>.

e. tā (**zuì-duō**) huā-le [sān]$_F$ tiān **<u>jiù</u>** kàn-wán [yī]$_F$ běn shū le
3SG (at-most) take-PRF three day read-finish one CL book SFP
'It took her <u>no more than</u> three days to finish one book.' (cf. (2a), (2d))
<u>Less</u> time was spent <u>than expected</u>; <u>more</u> was accomplished <u>than expected</u>.

The examples in (1) and (2) do not exhaust all focus-sensitive scalar particles in Chinese, but even this brief glimpse of the data raises a series of research questions:

- How do we account for norm-sensitive inferences in the absence of gradable expressions (e.g., the gradable adjective *tall* in *she is tall*)?
- How is information projected from the focus associate?
- The focus associate can occur to the right or left of the particle (see (1a) vs. (1b), (2a) vs. (2d)/(2e)). Do these different positions affect sentence interpretation?
- When multiple particles co-occur (e.g., (1b), (1c), (2d), (2e)), is there a division of labor, and how do we explain their optional or obligatory presence?
- Particles such as *dōu*, *cái*, and *jiù* are multi-functional. How does their norm sensitivity shed light on a unified understanding of their various uses?

This chapter addresses these questions by focusing on ***dōu***, ***cái***, and ***jiù***, three representative particles illustrated in (1) and (2). Building on Zhang (2022)'s analysis of English *even*, I argue that the key function of these particles is to invoke a contextually salient degree QUD, which their prejacent resolves with maximal informativeness, thereby inducing norm-sensitive inferences (§1.1). Based on this analysis, I examine the different monotonicity patterns of information projection in the use of *dōu*/*cái*/*jiù* (§1.2). Then I discuss the positions of their focus associates (§1.3), the division of labor and collaboration among particles (§1.4), and the multi-functionality of *dōu*, *cái*, and *jiù* (§1.5). §1.6 concludes this chapter.

This chapter is largely informal and non-technical, and it does not aim for empirical or theoretical exhaustiveness. I hope to show (i) how a degree-QUD-based perspective bridges sentence-level truth conditions and discourse-level pragmatic inferences, and (ii) how cross-linguistic data refine and advance existing semantic theories.[1]

[1] In recent manuscripts (e.g., Zhang 2024, 2025, and Zhang (in prep.)), I provide more formal details and further discuss and compare the 'below-norm' particles in (2): *zhǐ*, *cái*, and *jiù*.

1.1 DERIVING THE NORM SENSITIVITY OF DŌU, CÁI, AND JIÙ

1.1.1 Zhang (2022): A degree-QUD-based perspective on English even

According to the canonical view (e.g., Horn 1969), English *even* brings two presuppositions: entity-based additivity (see (3a)) and likelihood-based scalarity (see (3b)).

(3) Even [Mary]$_F$ came.

a. (3) $\models$ Someone other than Mary came.

b. (3) $\models$ Compared to others, Mary was unlikely to come.

However, this view is challenged by the felicity of uttering (5) in the scenario in (4), where Eeyore was the only animal who took a bit of thistles and spit them out. Evidently, this felicitous use of *even* is based on neither additivity nor low likelihood.

(4) Scenario (see Szabolcsi 2017): Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known to favor thistles) takes a bite but spits it out.

(5) (Those thistles must be really prickly!) Even [Eeyore]$_F$ spit them out!

a. (5) $\not\models$ Someone other than Eeyore spit the thistles out.

b. (5) $\not\models$ Compared to others, Eeyore was unlikely to spit the thistles out.

Zhang (2022) points out that in view of question-answer congruence, the *even*-sentence in (5) does not aim to address issues like *who spit the thistles out* or *who was unlikely to spit the thistles out*. Rather, *even* [*Eeyore*]$_F$ *spit the thistles out* provides information in resolving a degree QUD, such as *how prickly the thistles are*. Therefore Zhang (2022) proposes a degree-QUD-based perspective on the semantics of *even*:

(6) **The presuppositions of *even*(p)** (see also Greenberg 2018, Zhang 2025):

a. There is a contextually salient degree QUD.

b. Compared with its alternatives, the prejacent p is maximally informative in resolving the degree QUD.

Following Krifka (2000), Zhang (2025) argues that (6b) has a further consequence: the prejacent of *even*, p, resolves the degree QUD with **norm-sensitive effects**.

Suppose that among the animals, Pooh is also highly tolerant, perhaps at least as tolerant as Eeyore. However, as an alternative to the uttered proposition (here, *Eeyore spit the thistles out*), *Pooh spit the thistles out* would not be more informative in addressing the prickliness of the thistles. In other words, the information provided by *Eeyore spit the thistles out* yields a **superlative** value on the prickliness scale for thistles. Evidently, a superlative value is necessarily norm-sensitive: e.g., it cannot be the case that the most prickly thistles fail to reach the threshold for being prickly.

(7) shows another example. Here the focus associate of *even*, [3]$_F$, is maximally informative in addressing the relevant degree QUD. Thus, the use of a higher number would not lead to higher informativeness: From '3' onwards, the burden is always interpreted as maximally heavy (see also Rullman 1997).

(7) (QUD: *how heavy is the burden of parenting?*) Bill even has [3]$_F$ kids.

1.1.2 A degree-QUD-based perspective on Chinese dōu, cái, and jiù

I propose that, like English *even*, Chinese *dōu/cái/jiù* invoke a contextually salient degree QUD, which their prejacent resolves with **maximal informativeness**. Using a **bare numeral** in *dōu*(p)/*cái*(p)/*jiù*(p) reveals how the prejacent p is differently positioned among the alternatives in addressing such a degree QUD (to be revised):

(8) a. *dōu*(p): p corresponds to the **maximal** degree (by reaching it)
  b. *cái*(p)/*jiù*(p): p corresponds to the **minimal** degree (by not exceeding it)

As illustrated in (9), *dōu/cái/jiù* invokes a salient degree QUD and QUD-related presuppositions (see (6)). The accommodation of these presuppositions imposes **constraints and an ordering** on the alternative set of the focus associate. The *dōu*-sentence in (9a) means that she has **reached the maximal degree**, i.e., the expected alternatives fall below the uttered value, 3. Thus she is older than expected. In contrast, the sentences in (9b) mean that she **does not exceed the minimal degree**, i.e., she does not exceed any of the expected alternatives above 3. Thus she is not as old as expected. Overall, we interpret (9a) and (9b) with norm-sensitive inferences.

(9) a. tā **<u>dōu</u>** [sān]$_F$ suì le
  3SG EVEN three year SFP
  'She is already 3.' (=(1a)) ***dōu*** ≈ *even*, *already*, German *schon*
  (QUD: *how old is she?*) The alternative set: $\{..., \geq 1$ yo$, \geq 2$ yo$, \underline{\geq 3 \text{ yo}}\}$
 b. tā { **<u>cái</u>** / **<u>jiù</u>** / **<u>zhǐ</u>** yǒu } [sān]$_F$ suì
  3SG JUST JUST ONLY have/EXIST three year
  'She's just 3.' (=(2a)) ***cái/jiù/zhǐ*** ≈ *only*, *no more than*, German *erst*
  (QUD: *how old is she?*) The alternative set: $\{\underline{\leq 3 \text{ yo}}, \leq 4$ yo$, \leq 5$ yo$, ...\}$
  i.e., {<u>She is **not** older than 3</u>, She is **not** older than 4, ...} (Zhang 2025)

1.1.3 The role of a contextually salient degree QUD

It is worth noting that, in interpreting *dōu/cái/jiù*-sentences, the ordering of the alternative set is determined not by the numerical value of the focus associate itself, but rather by **informativeness** relative to the **contextually salient degree QUD**.

For both (9a) and (9b), the interpretation involves degree QUDs anchored to the direction of age increase: for *dōu*-sentence (9a), higher numerical values correspond to greater age and hence higher informativeness (e.g., '≥ 3 yo' is more informative than '≥ 2 yo'); whereas for sentences in (9b), the negation of lower numerical values corresponds to higher informativeness (e.g., negating '> 3' is more informative than negating '> 4'). In other contexts, the relevant QUD may impose a different ordering.

As illustrated in (10) (see Fig. 1.1 and 1.2), in a **growth** context, higher numerical weight values correspond to greater growth and higher informativeness. Thus, in interpreting the *dōu*-sentence in (10a), the focus associate of *dōu*, 70 kg, is the largest numerical value in the alternative set, yielding a 'heavier than expected' inference. In contrast, in the same context, in interpreting the sentences in (10b), the focus associate of *cái/jiù/zhǐ*, 70 kg, involves the smallest numerical value in the alternative set, resulting in a 'not as heavy as expected' inference.

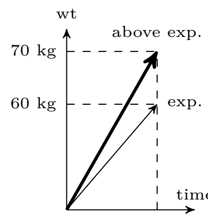

Figure 1.1
tā dōu [70]$_F$ *kg le*
(Growth)

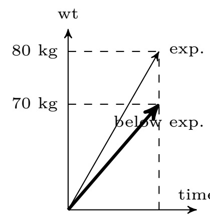

Figure 1.2
tā cái [70]$_F$ *kg*
(Growth)

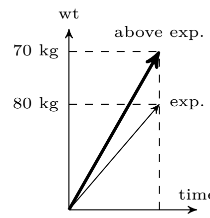

Figure 1.3
tā dōu dào [70]$_F$ *kg le*
(Weight-loss)

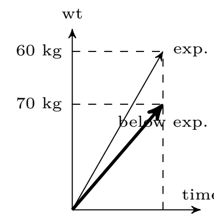

Figure 1.4
tā cái dào [70]$_F$ *kg*
(Weight-loss)

(10) **Growth context**

a. tā **dōu** [qī-shí]$_F$ qiān-kè le, (zhǎng-de hǎo kuài)
3SG EVEN seven-ten kilo-gram SFP grow-LINKER really fast
'He even weighs 70 kg. (He is growing so fast.)'
70 is the largest numerical value: {... ≥ 50 kg, ≥ 60 kg, ≥ 70 kg}
⇝ **exceeding expected heaviness (or growth)** (see Fig. 1.1)

b. tā **cái/jiù/zhǐ**yǒu [qī-shí]$_F$ qiān-kè, (zhǎng-de hǎo màn)
3SG JUST/ONLY seven-ten kilo-gram grow-LINKER really slow(ly)
'He just weighs 70 kg. (He is growing so slowly.)'
70 is the smallest numerical value: {≤ 70 kg, ≤ 80 kg, ≤ 90 kg, ...}
i.e., {He is **not** heavier than 70 kg, He is **not** heavier than 80 kg, ...}
⇝ **below expected heaviness (or growth)** (see Fig. 1.2)

On the other hand, as illustrated in (11) (see Fig. 1.3 and 1.4), in a **weight-loss** context, lower numerical weight values correspond to greater weight loss and higher informativeness. Thus, in interpreting the *dōu*-sentence in (11a), the focus associate of *dōu*, 70 kg, is the smallest numerical value in the alternative set, yielding a 'lighter than expected' inference. In contrast, in the same context, in interpreting the sentences in (11b), the focus associate of *cái/jiù/zhǐ*, 70 kg, is the largest numerical value in the alternative set, resulting in a 'not as light as expected' inference.

(11) **Weight-loss context**

a. tā **dōu** dào [qī-shí]$_F$ qiān-kè le, (shòu-le hǎo duō)
3SG EVEN arrive seven-ten kilo-gram SFP thin-PRF really much
'He has even reached 70 kg. (He has lost a lot of weight.)'
70 is the smallest numerical value: {... ≤ 90 kg, ≤ 80 kg, ≤ 70 kg}
⇝ **exceeding expected lightness (or weight loss)** (see Fig. 1.3)

b. tā **cái/jiù/zhǐ** dào [qī-shí]$_F$ qiān-kè, (xūyào jìxù jiǎn-féi)
3SG JUST/ONLY arrive seven-ten kilo-gram need continue lose-fat
'He has just reached 70 kg. (He needs to lose more weight.)'
70 is the largest numerical value: {≥ 70 kg, ≥ 60 kg, ≥ 50 kg, ...}
i.e., {He is **not** lighter than 70 kg, He is **not** lighter than 60 kg, ...}
⇝ **below expected lightness (or weight loss)** (see Fig. 1.4)

1.2 INFORMATION PROJECTION WITH DŌU, CÁI, AND JIÙ

1.2.1 Degree QUDs that involve single-dimensional information

Under the current analysis, *dōu/cái/jiù* are alike in (i) invoking a degree QUD and (ii) requiring their prejacent to be maximally informative (see (6)). However, the interpretation of bare numerals in *dōu/cái/jiù*-sentences reveals that *dōu* vs. *cái/jiù* differ in how maximal informativeness is encoded (see §1.1.2).

Given that *dōu* marks **reaching the maximal degree**, projection from the focus associate to sentence-level informativeness is **order-preserving** (monotone). By contrast, since *cái/jiù* mark **not exceeding the minimal degree**, information projection is **order-reversing** (antitone).

This projection difference is reflected in the compatibility of *dōu/cái/jiù* with different modified numerals. With a degree QUD like *how old she is*, *cái/jiù* are incompatible with upward-entailing (UE) numerals, but compatible with downward entailing (DE) numerals (see (12)), while *dōu* shows the opposite pattern (see (13)).[2]

(12) a. * tā cái/jiù zhì-shǎo [sān]$_F$ suì
3SG JUST at-least three year
Intended: 'She is at least 3 and not any more.' ***cái/jiù*+UE:** ×

b. tā cái/jiù zuì-duō [sān]$_F$ suì
3SG JUST at-most three year
'She is at most 3 and not any more.' ***cái/jiù*+DE:** ✓

(13) a. tā dōu zhì-shǎo [sān]$_F$ suì le
3SG EVEN at-least three year SFP
'She is at least 3.' ***dōu*+UE:** ✓

b. ? tā dōu zuì-duō [sān]$_F$ suì le
3SG EVEN at-most three year SFP
Intended: 'She is at most 3.' ***dōu*+DE: degraded**

The data and analysis so far are based on degree QUDs that hinge on a **single-dimensional** measurement (e.g., age, weight). Accordingly, (14) characterizes the required information projection in *dōu*- vs. *cái/jiù*-sentences, from the focus associate X to sentence-level informativeness measure $f_{\text{INFO}}(X)$:

(14) a. *dōu* (order-preserving): if $X \geq Y$, $f_{\text{INFO}}(X) \geq f_{\text{INFO}}(Y)$
b. *cái/jiù* (order-reversing): if $X \leq Y$, $f_{\text{INFO}}(X) \geq f_{\text{INFO}}(Y)$

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[2] The degradedness of (13b) largely stems from the presence of sentence-final particle (SFP) *le*. Roughly speaking, *le* is preferred in a 'reaching' event that has an endpoint or milestone (e.g., along the direction towards higher informativeness), but is incompatible with a 'not exceeding' event that lacks such an endpoint (see e.g., Lü 1980/1999). In (13b), *le* therefore clashes with *at most 3*.

Of course, a DE item can be embedded under *dōu* (see the discussion of the 'weight-loss' context in §1.1.3). Thus, both (ia) and (ib) are interpreted with the alternative set $\{\underline{< 3\text{ yo}}, < 4\text{ yo}, \ldots\}$.

(i) a. tā dōu bú-dào [sān]$_F$ sùi
3SG EVEN below three year
'She is not even 3.'

b. tā cái bú-dào [sān]$_F$ sùi
3SG JUST below three year
'She is not even 3.'

1.2.2 Degree QUDs that involve multi-dimensional information

Some linguistic constructions encode information across **multiple dimensions** to address a single underlying degree QUD, as illustrated in (15):

(15) a. At most 3% of the population owns (even) more than 70% of the land.
b. Fewer and fewer people own (even) more and more wealth.

For a **cumulative-reading** sentence like (15a), as noted by Krifka (1999) (see also Zhang 2023), maximal informativeness is not achieved by maximizing values along each dimension (here, population size and wealth amount). Otherwise, an alternative like *100% of the population owns 100% of the land* would be more informative. Instead, (15a) addresses such a degree QUD (e.g., *how skewed the wealth distribution is*): a **small** value on one scale (e.g., population), combined with a **large** value on another (e.g., wealth), jointly resolves the QUD with maximal informativeness.

Similarly, for a **multi-head comparative** like (15b), it is the interplay between a **decrease** on one scale (e.g., population) and an **increase** on another (e.g., wealth) that yields greater informativeness from the omitted *than*-clause to the matrix clause, relative to a degree QUD such as wealth imbalance (see Zhang 2024).

By integrating multi-dimensional information, the sentences in (15) resolve a degree QUD with maximal informativeness (see (16)). They satisfy the presuppositional requirements of *even* (see (6)), supporting its optional presence (Sigrid Beck, p.c.).

(16) For (15), the relevant degree QUD involves an f_{INFO} s.t. if $x_1 \leq x_2, y_1 \geq y_2$, then $f_{\text{INFO}}(x_1, y_1) \geq f_{\text{INFO}}(x_2, y_2)$ (here x_i: population; y_i wealth amount)

For the sentences in (15), although their degree QUD is implicit, the monotonicity property of the degree QUD's informativeness measure f_{INFO} (see (16)) is signaled by explicit clues: the opposing monotonicity of the numerals in (15a) (*at most* vs. *more than*), and the opposing polarity of changes in (15b) (*fewer* vs. *more*).

Intriguingly, in Chinese, *cái*- and *jiù*-sentences can also involve multi-dimensional information. As illustrated in (17) and (18), these cumulative-reading sentences support the presence of *cái* and *jiù*. The *cái*-sentence in (17) and the *jiù*-sentence in (18) share the same truth-conditional semantics, but differ in the patterns of modified numerals involved, leading to different sentence-level norm-sensitive emphases.

In (17), a UE numeral ('more than 500') precedes and a DE numeral ('fewer than 7') follows the use of *cái*, shifting from a large to a small quantity and emphasizing the paucity of books purchased. In contrast, in (18), a DE numeral precedes and a UE numeral follows the use of *jiù*, emphasizing the abundance of money spent.

(17)

tā	huā-le	chāo-guò	wú-bǎi	kuài	**cái**	mǎi-le	bú-dào	qī	běn	shū
3SG	spend-PRF	above	500	dollar		buy-PRF	below	7	CL	book

'She spent (even) more than 500 dollars on fewer than seven books.'

(18)

tā	mǎi-le	bú-dào	qī	běn	shū	**jiù**	huā-le	chāo-guò	wú-bǎi	kuài	le
3SG	buy-PRF	below	7	CL	book		spend-PRF	above	500	$	SFP

'She spent (even) more than 500 dollars on fewer than seven books.'

Thus, as characterized in (19), from a degree-QUD perspective, when integrating multi-dimensional information, *cái* and *jiù* both convey disproportionality, but differ in how informativeness is projected from multiple focus associates, resulting in slightly different sentence-level norm-sensitive emphases:

(19) a. *cái*: if $x_1 \geq x_2, y_1 \leq y_2$, then $f_{\text{INFO}}(x_1, y_1) \geq f_{\text{INFO}}(x_2, y_2)$
b. *jiù*: if $x_1 \leq x_2, y_1 \geq y_2$, then $f_{\text{INFO}}(x_1, y_1) \geq f_{\text{INFO}}(x_2, y_2)$

The current analysis accounts for seemingly inconsistent intuitions about the uses of *cái/jiù* in (2). In (2a)–(2c), *cái/jiù* bring a clear 'below-norm' inference, while in (2d) and (2e) they appear to give rise to mixed inferences: one part of the sentence yields an 'above-norm' inference, and another part yields a 'below-norm' inference. These intuitions precisely track the antitone (order-reversing) property of *cái/jiù*'s requirements (see (14b) and (19)): projection from exactly one focus associate to sentence-level informativeness is order-reversing (monotonically decreasing).

The current analysis also explains a widely discussed generalization: (*zhǐ-yǒu*) ... *cái* marks a necessary (and demanding) condition, while (*zhǐ-yào*) ... *jiù* marks a sufficient (and easy) condition (see (20) vs. (21); e.g., Hole 2004, Tsai 2017, Sun 2021, Wimmer 2022). According to (19a), projection from *cái*'s left-focus-associate is **order-preserving**, so in (20), 'five' is interpreted as '(need to be) at least five', exceeding expected alternatives and yielding the 'necessary and demanding' inference. By contrast, according to (19b), projection from *jiù*'s left-focus-associate is **order-reversing**, so in (21), 'five' is interpreted as 'not (need to be) above five' – not as high as expected alternatives, yielding the 'sufficient and low-demanding' inference.

(20) (**zhǐ-yǒu**) fā [wǔ]$_F$ piān lùn-wén, **cái** néng shēng [fù-jiào-shòu]$_F$
(only if) publish five CL paper can upgrade associate-prof
'One can be promoted to associate professor only if she has 5 publications.'

(21) (**zhǐ-yào**) fā [wǔ]$_F$ piān lùn-wén, **jiù** néng shēng [fù-jiào-shòu]$_F$
(only-need) publish five CL paper can upgrade associate-prof
'One can be promoted to associate prof. as long as she has 5 publications.'

1.3 THE POSITIONS OF THE FOCUS ASSOCIATES

The above discussion of cumulative-reading *cái/jiù*-sentences shows that projection from the left- vs. right-focus-associate of *cái/jiù* can exhibit different monotonicity.

What about *dōu*? Its focus associate can appear either to its right or to its left (see (1a) vs. (1b)). Y. Jiang (1998) notes that this position difference does not affect interpretation: (22) and (23) have identical meanings (see also e.g., Pan 2006).

(22) dōu (dào) [sān]$_F$ diǎn le. zěn-me hái méi xià kè?
EVEN arrive three o'clock SFP how-come still NEG end class
'It's already 3 o'clock. Why hasn't the class ended?'

(23) (lián) [sān]$_F$ diǎn dōu dào-le. zěn-me hái méi xià kè?
(EVEN) three o'clock EVEN arrive-PRF/SFP how-come still NEG end class
'It's already 3 o'clock. Why hasn't the class ended?'

Under the current analysis, *dōu* modulates QUD resolution at the sentence level (see (6)). Therefore, the position of the focus associate should not affect sentence interpretation. *dōu* typically precedes the predicate (and follows an overt topic/subject), so the linear order between *dōu* and its focus associate largely depends on whether, in addressing a salient QUD, the relevant alternative set is invoked from the predicate (to the right of *dōu*) or the topic/subject (to the left of *dōu*).

As illustrated in (24), focusing *Dan* in (24a) invokes an alternative set that ranks the kids. The prejacent, *Dan got 88*, is maximally informative regarding a degree QUD such as *how universally the kids performed well*, *how unchallenging the final was*. The utterance is felicitous only if Dan is ranked as the **weakest** kid.

By contrast, focusing *98* in (24b) invokes an alternative set regarding the grades. The prejacent, *Dan got 98*, involves the **highest** grade and is maximally informative in addressing a degree QUD like *how high the kids' performance can be*.[3]

(24) Context: A dad is talking about his kids: 'Overall, my four kids have satisfactory performance in the final. Ann got 95; Ben and Ken got 90; ...'

a. (lián) [Dan]$_F$ dōu dé-le bā-shí-bā
(EVEN) Dan EVEN receive-PRF eight-ten-eight
'Even Dan got 88.' {..., Ken's grade $\geq$ 88, Dan's grade $\geq$ 88}

b. Dan dōu dé-le [jiǔ-shí-bā]$_F$
Dan EVEN receive-PRF nine-ten-eight
'Dan even got 98.' {..., Dan's grade $\geq$ 95, ..., Dan's grade $\geq$ 98}

It is worth noting that, although the default word order in Modern Chinese is SVO, objects that serve as focus associates are often preferred in a fronted position to the left of *dōu*. Both (25a) and (25b) are grammatical and have the same meaning, but (25a) is intuitively preferred.

(25) a. tā (lián) [Faust]$_F$ dōu dú-le
3SG (EVEN) Faust EVEN read-PRF/SFP
'She even read *Faust*.'

b. tā dōu dú-le [Faust]$_F$ le
3SG EVEN read-PRF Faust SFP
'She even read *Faust*.'

[3]Under the context in (24), (ia) and (ib) are infelicitous, because the felicity requirements of *dōu* are not met. In (ia), the prejacent is not maximally informative in its alternative set, while in (ib), there is no salient degree QUD for the prejacent to resolve with maximal informativeness (i.e., the prejacent fails to correspond to a degree higher than the other alternatives).

(i) a. * Dan dōu dé-le [bā-shí-bā]$_F$
Dan EVEN receive-PRF eight-ten-eight
Intended: 'Dan even got 88.' {..., Dan's grade $\geq$ 88, ..., Dan's grade $\geq$ 95, ...}

b. * (lián) [Dan]$_F$ dōu dé-le jiǔ-shí-bā
(EVEN) Dan EVEN receive-PRF nine-ten-eight
Intended: 'Even Dan got 98.' {..., Ken's grade $\geq$ 98, Dan's grade $\geq$ 98}

This preference aligns with J. Jiang & Pan (2013): (i) when the invoked alternative set has a natural ordering (e.g., a set of numbers), the focus associate is preferred to stay in situ, i.e., after *dōu* (e.g., (24b); (22) is preferred to (23)); (ii) when the invoked alternative set lacks a natural ordering, the focus associate is preferred to be fronted to the left of *dōu* (e.g., (25a) is preferred to (25b)).

In (25), the alternative set invoked by the focus associate *Faust* (i.e., a set of books) lacks a natural ordering per se. Rather, any ordering arises from a salient degree QUD (e.g., *how wide her reading list is*, *how high her German reading ability is*) (Zhuang Chen, p.c.). Presumably, in the absence of a natural ordering, fronting the focus associate enhances its prominence beyond prosodic stress alone, facilitating a maximally informative interpretation. In addition to fronting, the optional particle *lián* can further enhance the prominence of the fronted focus associate in (25a).[4]

1.4 THE DIVISION OF LABOR AND COLLABORATION

Focus-sensitive scalar particles can co-occur. To convey the 'below-norm' meaning, in (26) and (27), 7 and 3 particle combinations are acceptable, respectively.[5]

(26) **7 acceptable combinations, with no meaning distinction:**
jiù, cái, zhǐ, jiù zhǐ, cái zhǐ, jiù cái, jiù cái zhǐ

tā (**jiù**) (**cài**) (**zhǐ** yǒu) [sān]$_F$ suì
3SG (JUST) (JUST) (ONLY have/EXIST) three year

'She is only/just 3 years old.'

(27) a. Zhāng-Sān (**jiù**) (**zhǐ**) chī-le [píng-guǒ]$_F$
Zhāng-Sān (JUST) (ONLY) eat-PRF apple(s)
'Zhāng Sān only ate apples.' **3 possibilities: *jiù, zhǐ, jiù zhǐ***

b. (**jiù**) (**zhǐ** yǒu) [Zhāng-Sān]$_F$ lái-le
(JUST) (ONLY have/EXIST) Zhāng-Sān come-PRF/SFP
'Only Zhāng Sān came.' **3 possibilities: *jiù, zhǐ, jiù zhǐ***

In their multi-dimensional use, *cái/jiù* are obligatory, while all parenthesized items are optional and may co-occur, yielding 4 possibilities for each sentence in (28).

(28) a. tā (**zhǐ-yǒu**) (**zhì-shǎo**) huā [sān]$_F$ tiān **cái** néng kàn-wán
3SG (only if) (at-least) take three day can read-finish
'She needs no less than three days to finish reading.'

b. tā (**zhǐ-yào**) (**zuì-duō**) huā [sān]$_F$ tiān **jiù** néng kàn-wán
3SG (only-need) (at-most) take three day can read-finish
'She needs no more than three days to finish reading.'

[4]For the single-dimensional use of *cái* and *jiù* (see §1.2.1), the focus associate is always to their right. Presumably, the left side of *cái/jiù* is reserved for the multi-dimensional use (see §1.2.2).

[5]In its single-dimensional use, *cái* requires its focus associate's alternative set to have a natural ordering (e.g., numbers, rankings). Thus, *cái* can only be used in cases like *she is only* [*3*]$_F$ (see (26)), *she only got* [*B+*]$_F$. In (26)/(27), when *zhǐ* 'only' precedes a DP, verb *yǒu* must be inserted; this is distinct from the particle *zhǐ-yǒu* 'only if', which is often used along with *cái* (see (28a)).

In (29), *dōu* is obligatory for a scalar meaning, while all parenthesized items are optional and may co-occur, yielding 4 possibilities. A scalar reading of the *yě*-sentence (30) needs at least one of the two particles, *shèn-zhì* or *lián*, yielding 3 possible combinations. In (31), where the focus associate is from the predicate, at least one of the two particles, *shèn-zhì* or *dōu*, is needed, yielding 3 possibilities.

(29) **(shèn-zhì)** **(lián)** [Zhāng-Sān]$_F$ **dōu** lái-le
(EVEN) (EVEN) Zhāng-Sān EVEN come-SFP
'Even Zhāng Sān came.' **4 acceptable combinations**

(30) { **shèn-zhì** / **lián** } [Zhāng-Sān]$_F$ **yě** lái-le
EVEN EVEN Zhāng-Sān ALSO come-SFP
'Even Zhāng Sān came.' **3 acceptable combinations**

(31) tā (**shèn-zhì**) (**dōu**) dé-le [yī-bǎi]$_F$ fēn
3SG (EVEN) (EVEN) receive-PRF one-hundred point
'She even got 100 points.' **3 acceptable combinations**

Liao & Jheng (2025) make a distinction between two *even*-like particles in Chinese: *shèn-zhì* is clause-level, triggering a scalar inference, while *lián … dōu* is phrase-level, triggering an additive inference. However, it is unclear why there should be a correlation between syntactic units (clause vs. phrase) and inferences (scalar vs. additive). It is also questionable to treat *lián … dōu* as a single unit, since *dōu* is often used alone, while *lián* requires the presence of *dōu* or *yě* (see (29) and (30)).

From a degree-QUD perspective, the interpretation of the relevant data depends on two key elements: a salient degree QUD and a focus associate. The degree QUD imposes an **ordering** on the focus associate's alternative set (or makes use of its existing ordering), so that the focus associate represents maximal informativeness.

In English, these elements are often overtly realized by *even* (which invokes a degree QUD) and by prosodic stress (which marks the focus associate). Together, they suffice to derive a scalar, norm-sensitive (mirative) interpretation.

However, as shown by cumulative-reading sentences and multi-head comparatives (§1.2.2), the interplay of the monotonicity of numerals (or the polarity of changes) naturally gives rise to ordering and hints at a relevant degree QUD. In such cases, the overt presence of *even* is unnecessary (or it can be covertly accommodated).

Similarly, as illustrated in (32), minimizer NPIs (e.g., *a single person*) naturally induce an ordered set of alternatives, allowing mirativity to be expressed without an overt *even* (see Zhang 2025).

(32) Context: John was mentally withdrawn and closed off in his own mind.
The party was crowded, but John didn't (even) see a single person.
⇝ maximally informative in addressing *how mentally withdrawn John was*

Thus, I propose that focus-sensitive scalar particles fall into two categories (see (33)): (i) *even*-like particles that invoke a degree QUD, ordering the focus associate's alternative set by informativeness; and (ii) particles that mark extreme cases in an ordered alternative set. The latter helps highlight ordering, while the former invokes a degree QUD and thereby imposes ordering.

(33) a. Invoking a degree QUD and enforcing maximal informativeness:
• ***dōu***: with order-preserving projection
• ***jiù***, ***cái***: with order-reversing projection

b. Marking extreme cases in an ordered set:
• For an ordered set like $\{\leq x_1, \leq x_2, ...\}$ (where $x_1 < x_2 < ...$):
zhǐ 'no more than', ***zhǐ-yào*** 'not need more than', ***zuì-duō*** 'at most';
• For an ordered set like $\{..., \geq x_2, \geq x_1\}$ (where $x_1 > x_2 > ...$):
zhǐ-yǒu 'only if, need at least', ***zhì-shǎo*** 'at least';
• Marking the last item(s) along a scale or in the summation (of entities):
lián 'to the extent', ***shèn-zhì*** 'moreover', ***hái*** 'still'

Obviously, by invoking a degree QUD (and thereby imposing an ordering), the use of a particle in (33a) is usually sufficient and effective. Co-occurrence of multiple particles from (33) is allowed when their contributions converge (see e.g., (26), (27)).

When the focus associate's alternative set lacks a natural ordering, additional prominence is often needed to mark the focus associate as extreme, favoring co-occurrence (e.g., *lián ... dōu*, *zhǐ-yǒu ... cái*, *zhǐ-yào ... jiù*) and fronting.

Some particles from (33b), particularly *zhǐ* 'only, no more than' and *zuì-duō* 'at most', are often used independently. By marking the minimal value in a set $\{\leq x_1, \leq x_2, ...\}$, they naturally induce an order-reversing projection, providing clues to a relevant degree QUD, supporting a covert accommodation of *cái/jiù*, and yielding a scalar, norm-sensitive interpretation – reminiscent of minimizer NPIs (see (32)).[6]

In expressing a scalar reading, *lián* depends on *dōu/yě*. In (29), its optional presence enhances the prominence of the focus associate.

In (30), additive particle *yě* introduces an additive presupposition, making its focus associate invoke a non-singleton alternative set. By marking the extreme case of this set, *lián/shèn-zhì* bring an ordering (effectively supporting a covert *dōu*), yielding a scalar interpretation.[7]

Finally, (34) illustrates how particles collaborate. Here, *lián ... dōu* and *cái* bring an 'above-norm' inference (e.g., a reader with high proficiency) and a 'below-norm'

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[6] A similar idea can be adopted to analyze *hái*-containing comparatives (see (1d), repeated in (i); see also Chen 2025). In a comparative, the gradable adjective provides a scale. By marking its focus associate as an extreme case, *hái* restricts the invoked alternative set, supporting a covert accommodation of *dōu* and yielding a norm-sensitive interpretation.

This analysis can also be extended to account for the use of *shèn-zhì* in (31): its focus associate, *100 points*, makes an ordered alternative set readily available, supporting a hidden *dōu*.

(i) Zhāng-Sān bǐ [hóu-zi]$_F$ (**dōu**) **hái** mǐnjié
Zhāng-Sān COMPARE monkey (EVEN) EVEN agile
'Zhāng Sān is even more agile than a monkey.'
Alternative set: {... more agile than a panda, more agile than a monkey}

[7] It is worth noting that *shèn-zhì* often patterns with cross-domain scalar-additive (incremental) particles (e.g., *hái*), marking one of the last items in the summation of events or entities. Thus, *shèn-zhì ... yě* (see (30)) is ambiguous between 'moreover, Zhāng Sān also came' and 'even Zhāng Sān came'. In the former, the prejacent, *Zhāng Sān also came*, is not necessarily norm-sensitive; rather, including this event renders the entire sum of events 'above norm'. A detailed study of incremental particles is for another occasion.

inference (e.g., a small number of books), jointly addressing an underlying degree QUD (e.g., to what extent reading is a rare activity among people). Each particle plays its own role, and together their contributions align seamlessly.

(34) **<u>lián</u>** [Zhāng-Sān]$_F$ **<u>dōu</u>** **<u>cái</u>** **<u>zhǐ</u>** kàn-le [liǎng]$_F$ běn shū
EVEN Zhāng-Sān EVEN only read-PRF two CL book
'Even Zhāng Sān read only two books.'
(*lián*: optional; *dōu*: obligatory; *cái/zhǐ*: at least one required)

1.5 OTHER USES OF DŌU, CÁI, AND JIÙ

Beyond their role as focus-sensitive scalar particles, *dōu/cái/jiù* have other uses. I show that norm sensitivity remains a useful lens for a unified understanding: *dōu* signals a high degree along a scale or significant homogeneity among items, while *cái* and *jiù* both signal small values along a scale, but in opposite directions.

1.5.1 Chinese dōu ≈ English all

As illustrated in (35), Chinese *dōu* has a use closely resembling English *all*, involving definiteness and distributivity. Thus, its core semantic contribution has been characterized as universal quantification (e.g., Y. Jiang 1998, Pan 2006), distributivity (Lin 1998), significant extent (Xu 2007), maximization (Xiang 2008), etc. Recently Liu (2023) proposes that *dōu* marks a QUD-based informativeness maximality: the prejacent entails all alternative presuppositions.

(35) sān gè háizi **<u>dōu</u>** chī-le wǔ pán pīsà
three CL kid ALL eat-PRF five CL pizza
'<u>The</u> three kids <u>all/each</u> had five pizzas.' **Definiteness, distributivity**
(Without *dōu*: three kids had five pizzas. ✓ **distributive**, ✓ **cumulative**)

However, most existing theories (except Xu 2007) struggle to explain the contrast in (36): *dōu* can felicitously describe what is homogeneously shared among a large proportion of a given set (see (36a)), but is infelicitous when only a small subset is included (see (36b)). Contextually adjusting the relevant alternative set (while adopting entailment-based informativeness) (e.g., Liu 2023) cannot solve this issue.

(36) QUD: Among those 10 people, who came?
a. nà shí gè rén lǐ, chú-le liǎng gè, **<u>dōu</u>** lái-le
DEM ten CL human among except two CL ALL come-PRF/SFP
'Among the 10 people, all but two came.'
b. ? nà shí gè rén lǐ, chú-le bā gè, **<u>dōu</u>** lái-le
DEM ten CL human among except eight CL ALL come-PRF/SFP
Intended: 'Among the 10 people, all but eight came.'

Therefore, the *all*-like use of *dōu* is also norm-sensitive. A unified account for *dōu*'s uses should be grounded in a degree-QUD-based notion of informativeness and an 'above-norm' inference across domains of entities and degrees.

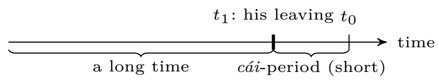

Figure 1.5 *tā cái zǒu* 'He just left' (reference time: t_0; his leaving time: t_1)

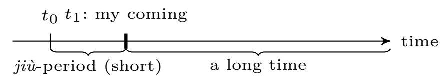

Figure 1.6 *wǒ jiù lái* 'I am coming soon' (reference time: t_0; my coming time: t_1)

1.5.2 Temporal uses of cái and jiù

Both *cái* and *jiù* have a temporal use: *cái* means 'just, a short while ago' (see (37) and Fig. 1.5), while *jiù* means 'right away, soon, shortly after' (see (38) and Fig. 1.6).[8]

(37) tā **cái** zǒu
3SG just leave
'He just left.' ('He left a short while ago.') (Fig. 1.5)

(38) wǒ **jiù** lái
1SG soon come
'I am coming soon.' (Fig. 1.6)

The temporal uses of *cái* and *jiù* show that both particles encode a short time interval away from a reference time t_0: before t_0 in the case of *cái*, and after t_0 in the case of *jiù*. As illustrated in Fig. 1.5 and 1.6, these temporal uses give rise to two inferences. First, the felicity of using *cái/jiù* increases as t_1 (i.e., the event time) approaches the reference time t_0. Second, the time interval preceding the *cái*-period and that following the *jiù*-period are inferred to be long.

These temporal uses therefore illuminate the projection patterns associated with *cái/jiù*. Their single-dimensional uses pattern with the inference observed for *cái/jiù*-periods: lower values correspond to higher informativeness, yielding an order-reversing projection. In their multi-dimensional uses, *cái/jiù* relate a large and a small value, encoding disproportionality. For *cái*, the large value precedes the small one, while for *jiù*, the small value precedes the large one. Overall, both the temporal and the focus-sensitive uses of *cái/jiù* rely on the same information projection mechanisms, differing only in the type of scales involved (the temporal timeline vs. other scales).

1.6 CONCLUSION

Based on a degree-QUD perspective, this chapter analyzes Chinese *dōu*, *cái*, and *jiù* as *even*-like particles. Under this proposal, the two elements identified in the canonical analysis of *even*—a likelihood scale and entity-based additivity—are replaced by a degree-QUD-based informativeness scale and additivity along this scale, both reflected in the norm-sensitive inferences of these particles.

This chapter demonstrates how this theoretical perspective yields new empirical insights and how cross-linguistic data contribute to theory development.

[8] Intriguingly, English focus-sensitive particle *just* also has this temporal use. For these temporal uses of *cái/jiù*, as well as the *all*-like use of *dōu*, prosodic stress falls on the particles themselves.

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